Scale or Shape? Decomposing NTK Movement During Grokking

Group G2
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Notes for the group. Delete this block before submission.

- **[TODO: Later, replace every mention of "rotation" with "shape", including R_t's name, the figure labels and the tier titles. The word "rotation" is kept for now so that the text matches what the group already uses.]**
- Each section opens with the claim it has to support. Every number comes from the scripts in `report_v2/code/` through `numbers.tex`. The only red planning values left are those of E14 in Appendix D.
- The main text must end by page 12. References and appendices do not count, and the appendices may take at most 5 pages. Each section shows its page budget.
- The draft notes after the consent page hold the criteria self-assessment, the changelog of what was cut and why, and the reference checks.

Abstract

Grokking is delayed generalisation, and what sets the length of the delay is disputed. Recent accounts time it by the weight norm, by the scale of the logits, or by the weight-decay rate. We split the movement of the neural tangent kernel into a change of scale and a rotation, and measure both, with the kernel's alignment to the labels, on 720 runs of a two-layer network learning modular addition. The runs cross laziness, width and weight decay. The usual distance between the kernel and its initial value mostly records the growth of the kernel. At a fixed width, rotation is the statistic most tied to the grokking step, yet networks can reach that rotation without grokking. Across widths no statistic takes a fixed value at grokking. The grokking time grows with width and falls with weight decay, with exponents of $+0.24$ and -0.63, which neither a rotation clock nor a weight-decay clock predicts alone. Read before any run groks, rotation is the best single predictor of the grokking time on held-out settings, and scale adds information that rotation lacks. Each published clock therefore describes one part of what sets the grokking time.

1 Introduction

Page budget: 1.0 page.

Grokking is the delayed generalisation that Power et al. (2022) found on small algorithmic datasets. A network fits its training data early, and its test accuracy rises from chance long afterwards. Modular addition learned by a two-layer perceptron is a small instance of it. Full-batch gradient descent on the squared error groks there with no explicit regularisation (Gromov, 2023). What sets the length of the delay is an open question.

One account holds that grokking is a transition from lazy training to rich training (Kumar et al., 2024). In lazy training the network behaves as a kernel method with its initial tangent kernel. In rich training the kernel adapts to the task. The evidence that the kernel changes is a distance: the empirical tangent kernel stays close to its initial value until the training data are fitted, and moves far from it afterwards (Mohamadi et al., 2024). A distance does not say how the kernel changed. A kernel can grow in scale while its eigenvectors stay fixed. It can also rotate, so that its eigenvectors change.

The difference matters for the accounts of the grokking time. Recent work times grokking by the weight norm (Khanh et al., 2026), by the scale of the logits (Khanh, 2026), or by the

weight-decay rate (Kim, 2026; Pracher et al., 2026). The result that Kumar et al. (2024) cite for weight decay forcing the kernel to change is that of Lewkowycz and Gur-Ari (2021), in which the kernel only shrinks. None of these accounts measures whether the kernel rotates.

We ask what sets the grokking time: the shape of the kernel, its scale, or the weight-decay rate. Our hypothesis is that rotation sets it. We measure three statistics of the empirical tangent kernel along training: its scale S_t, its rotation R_t and its alignment A_t with the labels. We do so on a grid of 720 runs that crosses laziness, width and weight decay. Our contributions are as follows.

1. We split the movement of the kernel into a change of scale and a rotation, and use each one as a timing variable. On the baseline of Kumar et al. (2024), the usual distance is mostly a change of scale: at $\alpha = 1$ the scale carries 90% of D_t^2 at grokking.
2. We test whether grokking happens at a fixed value of each statistic. A placebo test reads each run at another run's grokking step, so a statistic that merely stays in a narrow band does not pass.
3. We cross width with weight decay and fit how the grokking time scales with each. A rotation clock and a weight-decay clock predict different exponents.
4. We test whether the statistics, read before any run groks, predict when it will. Unlike the settings that produced them, these statistics can be measured in any network.

We find that rotation matters most and is not enough on its own: the grokking time depends on both width and weight decay.

The works we cite each follow one quantity: the distance the kernel moves (Mohamadi et al., 2024), the weight norm (Khanh et al., 2026), the logit scale (Khanh, 2026), or the weight-decay rate (Kim, 2026; Pracher et al., 2026). This report separates the growth of the tangent kernel from its rotation, and sets a scale clock, a rotation clock and a weight-decay clock against each other on one grid. The split also bears on the evidence for the lazy-to-rich account. A kernel that only grows keeps its eigenvectors, so a large distance does not by itself show that features were learned.

2 Related Work

Page budget: 1.2 pages.

Grokking. Power et al. (2022) report grokking on small algorithmic datasets. Nanda et al. (2023) reverse-engineer the algorithm a small transformer learns for modular addition and split training into memorisation, circuit formation and clean-up. Gromov (2023) gives closed-form weights that solve modular addition with a quadratic activation, and observes grokking under full-batch gradient descent with no explicit regularisation.

Lazy and rich training. In the infinite-width limit the tangent kernel stays constant, so the network trains as a kernel method (Jacot, Gabriel, and Hongler, 2020). Chizat, Oyallon, and Bach (2020) show that lazy training is a matter of scale. Multiplying a model's output by a factor α brings its training closer to that of its linearisation as α grows. Kumar et al. (2024) use this factor to move a network between the two regimes, and propose that grokking is the transition from lazy to rich dynamics. Mohamadi et al. (2024) prove that a permutation-equivariant kernel method needs a constant fraction of the p^2 pairs to learn modular addition, and observe that the kernel moves far from its initial value only after the training data are fitted.

What sets the grokking time. Khanh et al. (2026) find that networks trained with weight decay grok when the weight norm reaches a critical value, and that clamping the norm above that value delays grokking exponentially. Khanh (2026) show that under cross-entropy the norm acts through the scale of the logits, and that under squared error it acts through a different route. Kim (2026, Cor. 5, Eq. 20, arXiv v1) derives, for linear models trained with heavy-ball momentum β and weight decay, a grokking time of about $(1-\beta)/(\eta\lambda)$ times a logarithm of an initial amplitude, and observes this scaling on modular addition. Pracher et al. (2026) predict, for homogeneous networks trained on the squared error with weight decay, that the grokking time is set by the product of the learning rate and the weight decay.

The kernel under weight decay. For a network trained by gradient flow with L_2 strength λ, the kernel evolves as

$$\begin{aligned} \frac{d\Theta_t(x,x')}{dt} &= -2(k-1)\lambda\,\Theta_t(x,x') + T_t(x,x') + T_t(x',x), \\ T_t(x,x') &= -\sum_{x''} \partial_{\mu\nu} f_t(x)\,\partial_\mu f_t(x')\,\partial_\nu f_t(x'')\,\ell'(x''), \end{aligned} \tag{1}$$

where k is the degree of homogeneity of the network, the sum runs over the training inputs x'' and repeated parameter indices, and ℓ' is the derivative of the loss with respect to the output (Lewkowycz and Gur-Ari, 2021, Eqs. S5 and S6, arXiv v2). The weight decay appears only in the first term, which rescales the kernel. Only T_t can rotate it. At initialisation T_0 has expectation of order $1/N$ in the width N, a result that Lewkowycz and Gur-Ari (2021) take from Dyer and Gur-Ari (2020). In the infinite-width limit the kernel therefore only decays, with fixed eigenvectors (Lewkowycz and Gur-Ari, 2021, Thms. 1 and 2). **[TODO: Dyer and Gur-Ari (2020) is named here only as Lewkowycz and Gur-Ari's source. Add it to ref.bib only after checking it on arXiv.]**

Predictions. Table 1 sets out what each account predicts for the three tests of this report. If rotation sets the clock and T_t stays of order $1/N$ along training, a wider network rotates more slowly and groks later at a fixed weight decay. The weight decay should then matter little. A weight-decay clock predicts the opposite. Each account was tested on its own, so these predictions have not been set against each other on one grid. The weight-decay accounts also leave open what sets the time without weight decay. Kim (2026, p. 3, arXiv v1) lists that case, and the creation of new directions by feature learning, among the things its mechanism does not explain. Rotation of the tangent kernel measures one form of such feature learning (our reading).

Table 1: What each account predicts for the three tests. The exponents are those of $\log T = c + a \log N + b \log(\eta\lambda)$. "Not stated" means that the source makes no claim. Entries marked "ours" are our reading of the source.

Account	Fixed at grokking (Tier 1)	Width exponent a (Tier 2)	Decay exponent b (Tier 2)	Early predictor (Tier 3′)
Rotation clock (our hypothesis)	R_t reaches a threshold	$a > 0$, since T_t shrinks with width (ours)	$b \approx 0$	R_t
Weight-decay clock (Kim, 2026; Pracher et al., 2026)	Not stated	$a \approx 0$; width can enter Kim's law only inside the logarithm (ours)	$b \approx -1$	$\eta\lambda$ alone
Norm or scale clock (Khanh, 2026; Khanh et al., 2026)	The weight norm reaches a critical value; in our terms S_t at grokking is fixed (ours)	Not stated	Not stated	S_t

3 Methods

Page budget: 1.8 pages.

3.1 Task, network and training

The task is modular addition with modulus $p = 23$, at the baseline of Kumar et al. (2024, App. 8.3, arXiv v3). Each pair (a, b) of residues is encoded as the concatenation $x_{a,b} \in \mathbb{R}^{2p}$ of the one-hot codes of a and b. Its target $y_{a,b} \in \mathbb{R}^p$ is the one-hot code of $(a + b) \bmod p$. The dataset holds all $p^2 = 529$ pairs. Each seed draws its own split, with 476 pairs for training and 53 for testing, and its own initial weights.

The network is a two-layer perceptron without biases,

$$f(x;\theta) = N^{-1/2} W^{(2)} \phi\big(D^{-1/2} W^{(1)} x\big), \tag{2}$$

with input dimension $D = 2p$, width N, weights $W^{(1)} \in \mathbb{R}^{N\times D}$ and $W^{(2)} \in \mathbb{R}^{p\times N}$ drawn from $\mathcal{N}(0,1)$, and the rectifier $\phi(h) = \max\{h, 0\}$. The factor $N^{-1/2}$ is the NTK parameterisation of Jacot, Gabriel, and Hongler (2020). Following Kumar et al. (2024), the trained predictor is $\tilde{f}(x;\theta) = \alpha[f(x;\theta) - f(x;\theta_0)]$ with learning rate $\eta = \eta_0/\alpha^2$. A larger α gives lazier training. The loss is the squared error averaged over the outputs and the training pairs. Training is full-batch gradient descent with weight decay, $\theta_{k+1} = (1 - \eta\lambda)\theta_k - \eta\nabla L(\theta_k)$. For plain gradient descent this is the same as L_2 regularisation (Kim, 2026). Table 2 lists every setting.

Table 2: Every setting of the runs. One setting of $(\alpha, N, \eta\lambda)$ is a cell, and each cell has 5 seeds.

Setting	Value
Modulus, pairs	$p = 23$, all 529 pairs; 476 for training and 53 for testing
Split and seeds	Model seed $s \in \{0, \dots, 4\}$; the split uses data seed $42 + s$
Network	Equation. (2), no biases, rectifier, weights from $\mathcal{N}(0, 1)$
Width N	$\{50, 100, 200, 400, 800, 1600\}$
Output scale α	$\{0.5, 1, 2\}$
Learning rate	$\eta = \eta_0/\alpha^2$ with $\eta_0 = 100$
Weight decay $\eta\lambda$	$\{0, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$
Optimiser and loss	Full-batch gradient descent, no momentum; mean squared error
Budget	200 000 steps
Checkpoints	307 per run: step 0, 120 log-spaced steps to 200 000, and every 1 000 steps
Saved weights	48 steps per run
Runs	720, all finished
Compute	18.9 h summed over runs; 2.7 h elapsed on 12 workers of one Mac

3.2 The grid

Each axis of the grid tests one account. The output scale α sets how lazy the training is (Chizat, Oyallon, and Bach, 2020). The width N sets the size of T_t in (1), and so how fast the kernel can rotate. The weight decay $\eta\lambda$ sets the clock of Kim (2026). The grid has $3 \times 6 \times 8 = 144$ cells and 720 runs, and every run finished.

3.3 Scale, rotation and alignment

We measure the tangent kernel of one scalar output, the sum of the logits divided by $\sqrt{p}$, $g(x;\theta) = p^{-1/2} \sum_c f_c(x;\theta)$. This is the pseudo-NTK of Mohamadi, Bae, and Sutherland (2022). On the 53 test pairs of a seed, the kernel at step t is the matrix with entries $[K_t]_{ij} = \langle \nabla_\theta g(x_i;\theta_t), \nabla_\theta g(x_j;\theta_t) \rangle$. The kernel of $\tilde{f}$ is α^2 times the kernel of f, so the ratios below do not depend on which of the two is used. With Y the one-hot targets of the test pairs and $C = I - \frac{1}{53}\mathbf{1}\mathbf{1}^\top$, we record

$$S_t = \frac{\|K_t\|_F}{\|K_0\|_F}, \qquad R_t = 1 - \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \, \|K_0\|_F}, \qquad A_t = \frac{\langle CK_tC,\, CYY^\top C \rangle_F}{\|CK_tC\|_F \, \|CYY^\top C\|_F}. \tag{3}$$

The scale S_t is 1 at initialisation, and a value above 1 means the kernel grew. The rotation R_t is the kernel distance of Fort et al. (2020). It is 0 at initialisation and does not change when the kernel is rescaled. The alignment A_t is the centred alignment of Cortes, Mohri, and Rostamizadeh (2024) with the label kernel $YY^\top$, and it also does not change when the kernel is rescaled.

The usual measure of kernel movement is the relative variation $D_t = \|K_t - K_0\|_F / \|K_0\|_F$ (Geiger et al., 2020). It is a function of the other two,

$$D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1, \tag{4}$$

(appendix A). A kernel that only grows has $R_t = 0$ and $D_t = |S_t - 1|$. A large D_t therefore does not show that the kernel rotated.

[TODO: Add a notation table here if the group wants one: S_t, R_t, A_t, T, t^*, a, b, with the value of each at initialisation and how to read it. It costs about a quarter of a page.]

3.4 Event times

The memorisation step of a run is the first checkpoint at which its training accuracy reaches 0.99. Its grokking step T is the first checkpoint at which its test accuracy reaches 0.95. Appendix F repeats the analysis at the levels 0.80 and 0.90. Both events are counted from step 0. A run that does not reach a level within 200 000 steps is kept, and its grokking step is recorded as censored at the budget. The events are read on the checkpoint grid, so each one is known to lie between two checkpoints. The gap to the checkpoint before is at most 9.7% of the grokking step and 10.9% of the memorisation step. We use accuracy rather than loss because the test loss levels off above zero after the test accuracy is perfect. In the baseline cell it stays between 0.014 and 0.027.

3.5 Analysis

Thresholds (Tier 1). For each statistic we take its value at the grokking step of every run that groks, and measure the spread across runs by the standard deviation (SD). A statistic can have a small spread because it stays in a narrow band all through training. The placebo test separates the two cases. It reads each run at the grokking step of another run, drawn by a random permutation with no fixed points, and measures the SD again. The tightening ratio is the median placebo SD over 2 000 permutations divided by the SD at the true events, and p is the share of permutations whose SD is no larger. A ratio near 1 means the statistic is not tied to the event. The SD carries the units of its statistic, so the comparison between statistics rests on the ratio and on p.

Scaling (Tier 2). We fit the log-normal accelerated failure time model

$$\log T = c + a \log N + b \log(\eta\lambda) + \gamma \log N \log(\eta\lambda) + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1), \tag{5}$$

by maximum likelihood. A censored run contributes the probability that its grokking step exceeds the budget (Kalbfleisch and Prentice, 2002). Both logarithms are centred at the middle of the grid, so a and b are the slopes there. The fits use lifelines (Davidson-Pilon, 2019). The intervals resample whole cells and then seeds within cells, so that they include the misfit of the cells about the fitted plane.

Prediction (Tier 3′). We read S_t, $\log R_t$ and A_t at $t^* = 2\,430$, a saved step before any run reaches the grokking level. Four model families map them to the grokking step: kernel ridge regression with a Gaussian kernel, a linear accelerated failure time model with interaction terms, gradient-boosted trees with the accelerated failure time objective (Chen and Guestrin, 2016), and a gated recurrent unit (Cho et al., 2014) that reads the statistics at the 25 saved steps up to t^*. Two scores are used. R^2 is the share of the spread in $\log T$ explained on held-out runs. The concordance index is the share of pairs of runs that a model puts in the right order, so that 0.5 is a coin toss (Harrell et al., 1982). The five seeds of a cell are close copies, so whole groups are held out: one cell at a time, which tests interpolation, or one value of α at a time, which tests extrapolation. Appendix E gives the settings of each model.

4 Experiments

4.1 Baseline: the reproduction of Kumar et al.

Page budget: 0.8 page.

Claim. At the baseline of Kumar et al. (2024) the network memorises within a few thousand steps and groks much later, and a lazier network groks later. Over this time the kernel grows several-fold while R_t stays below 0.09.

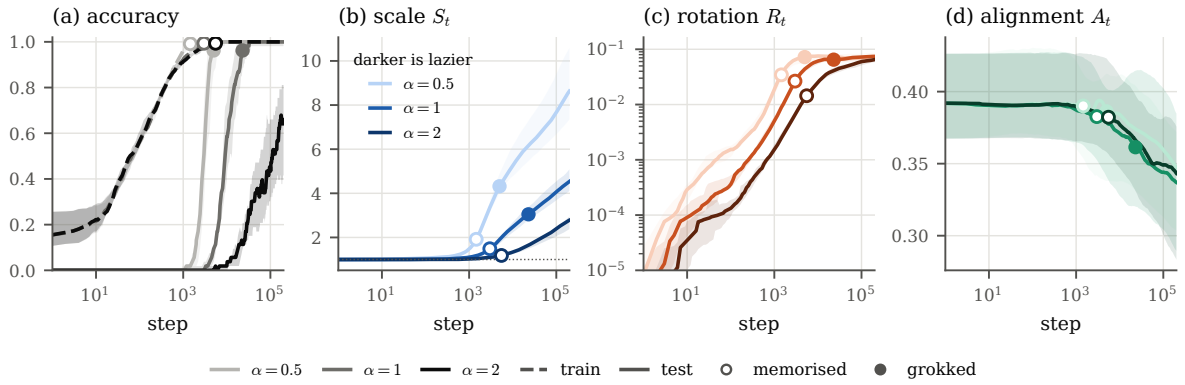


Figure 1: Training at the baseline of Kumar et al. (2024) ($N = 100$, $\eta\lambda = 0$) for three values of α. Each line is the median over 5 seeds and the shading is their range. Darker lines are lazier networks. A hollow marker is the median memorisation step and a filled marker the median grokking step. No seed groks at $\alpha = 2$. R_t is on a log scale.

At $\alpha = 0.5$ and $\alpha = 1$ every seed memorises and then groks (Figure 1). The median memorisation step is 1 455 at $\alpha = 0.5$ and 2 983 at $\alpha = 1$. The median grokking step is 4 981 (range 3 662 to 6 776) at $\alpha = 0.5$ and 23 000 (range 15 395 to 34 000) at $\alpha = 1$. At $\alpha = 2$ the network memorises by a median step of 5 520, but no seed reaches test accuracy 0.95 within 200 000 steps. Grokking therefore comes later as α grows, as in Kumar et al. (2024, Fig. 2(c), arXiv v3). The grokking steps at $\alpha = 0.5$ and $\alpha = 1$ are close to those in their figure, while our networks fit the training pairs several times sooner. They do not state their parameterisation or initialisation, which may explain the gap. The reproduction therefore matches on the grokking time and on the order in α, and does not match on the memorisation time. Appendix B compares the settings one by one.

By (4) the squared variation splits into a scale part $(S_t - 1)^2$ and a rotation part $2S_tR_t$. At $\alpha = 1$ the scale part is 76% of D_t^2 at memorisation, 90% at grokking and 95% at the last step. At the last step S_t is 4.5, while R_t is only 0.074. The scale part is 94% at grokking for $\alpha = 0.5$, and 90% at the last step for $\alpha = 2$. The distance D_t thus mostly records the growth of the kernel, which is why we measure S_t and R_t separately.

4.2 Tier 1: is there a threshold?

Page budget: 1.6 pages.

Claim. At a fixed width, rotation is the statistic most tied to the grokking step, but reaching that rotation does not make a network grok. The alignment at grokking barely differs from its value at initialisation. Across widths, no statistic takes a fixed value at grokking.

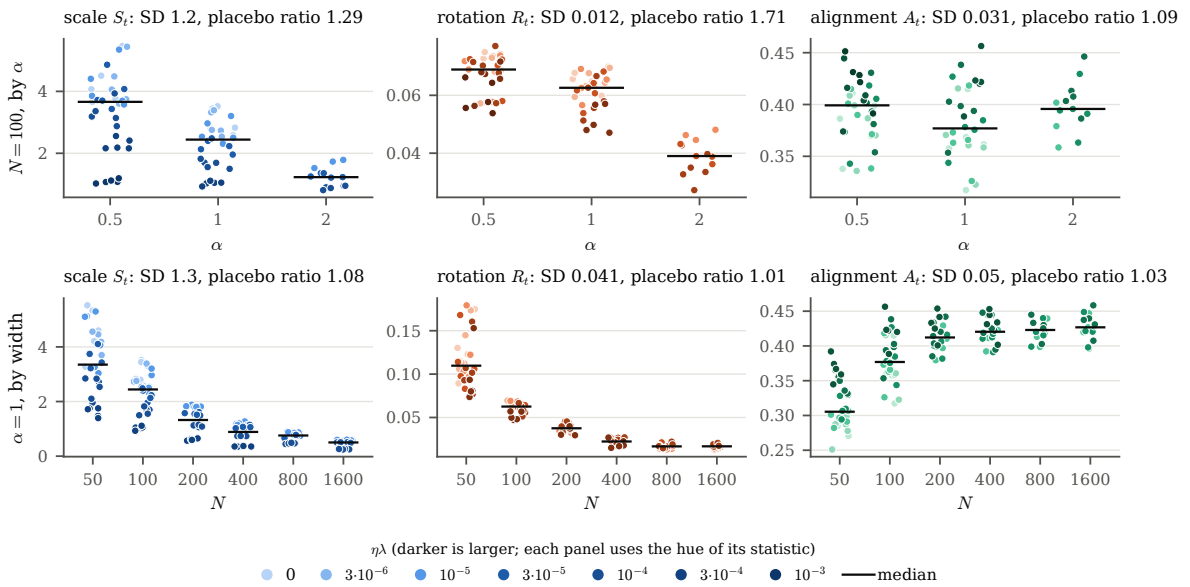


Figure 2: The kernel statistics at the grokking step of every run that groks. Top row: $N = 100$, grouped by α. Bottom row: $\alpha = 1$, grouped by width. Colour is $\eta\lambda$, darker for larger, and bars are medians. Each title gives the SD at grokking and the tightening ratio of the placebo test (Table 3).

Table 3: Spread of each statistic across runs that grok, as a standard deviation, at the grokking step, at the memorisation step and at another run's grokking step (placebo, median over 2 000 permutations). The ratio is the placebo SD over the SD at grokking, and p is the share of permutations with an SD no larger than the true one.

Runs	Statistic	At grokking	At memorisation	Placebo	Ratio	p
$N = 100$ (78 runs)	S_t	1.2	0.42	1.6	1.29	< 0.0005
	R_t	0.012	0.0094	0.02	1.71	< 0.0005
	A_t	0.031	0.02	0.034	1.09	0.029
$\alpha = 1$ (130 runs)	S_t	1.3	0.46	1.4	1.08	0.258
	R_t	0.041	0.035	0.041	1.01	0.423
	A_t	0.05	0.033	0.052	1.03	0.079

At $N = 100$, R_t is the statistic most tied to the event. Its SD widens 1.71-fold when each run is read at another run's grokking step, against 1.29-fold for S_t and 1.09-fold for A_t (Table 3). No permutation gave R_t or S_t a spread as small as the true one. The alignment tightens only a little ($p = 0.029$). Its median over the runs that grok is 0.398 at initialisation and 0.406 at grokking. Across widths no statistic passes the placebo test at level 0.95. At $\alpha = 1$ the ratios are 1.08, 1.01 and 1.03, and no p is below 0.05. At grokking R_t falls from 0.108 at $N = 50$ to 0.019 at $N = 1600$, and S_t falls from 4.11 to 0.58 (medians over all α and $\eta\lambda$). At level 0.80 (appendix F) the scale and the alignment tighten slightly across widths, and at 0.90 only the alignment does. R_t does not at any level. Pooled over all 355 runs that grok, S_t tightens 1.69-fold ($p < 0.0005$). The grokking steps of that pool span nearly two decades while S_t keeps growing, so this may reflect growth more than a threshold.

Reaching the rotation at which runs grok is not enough to grok. There are 34 cells in which some seed memorises but no seed groks. In 8 of them, at least one seed reaches the median R_t

at grokking of its width before the budget runs out. One of them is the baseline cell at $\alpha = 2$ (section 4.1).

Wider networks grok with less rotation but take longer, which suggests that width sets the rate at which the kernel rotates. Section 4.3 tests this. Our proposal expected the alignment to cross a threshold shared by every cell, and that does not hold. Other claims of a fixed value at grokking, such as the critical weight norm of Khanh et al. (2026), can be given the same placebo check.

4.3 Tier 2: width or weight decay?

Page budget: 2.0 pages.

Claim. The grokking time depends on both width and weight decay, so neither the rotation clock nor the weight-decay clock of Table 1 holds on its own. Width sets the rotation of the kernel. At a fixed width, weight decay shrinks its scale.

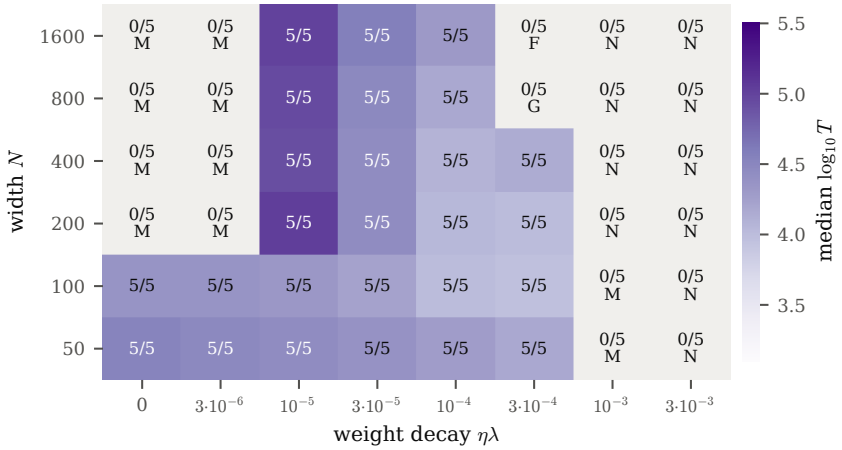


Figure 3: Grokking time and outcome over width and weight decay at $\alpha = 1$. Colour is the median $\log_{10} T$ over the seeds that grok, shown when at least 3 of 5 grok. Each cell gives the seeds that grok out of 5. A cell in which none grok also shows the most common phase at the last step: M memorises only, F forgets, N never fits, and G ends above test accuracy 0.9 without reaching 0.95.

Without weight decay, only $N = 50$ and $N = 100$ grok within the budget (Figure 3). The wider networks memorise and stop there. From $\eta\lambda = 10^{-5}$ to 10^{-4} every seed at every width groks, and the time falls with $\eta\lambda$ at every width. The largest weight decay at which every seed groks is 3×10^{-4} for $N \leq 400$ and 10^{-4} for $N = 800$ and $N = 1600$. At 10^{-3} and above no width groks. A wider network thus groks later at the same weight decay and stops grokking at a smaller one, so the weight decay that speeds grokking depends on the width.

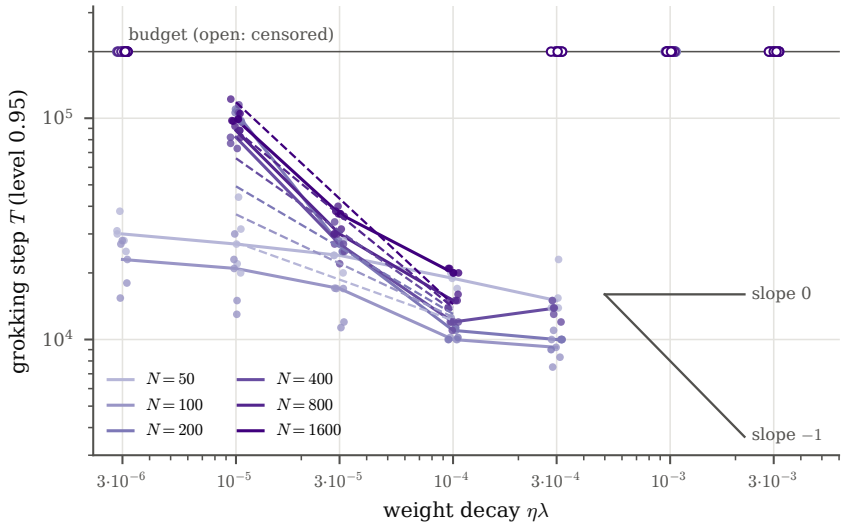


Figure 4: How the grokking time scales with weight decay at each width, at $\alpha = 1$. Points are seeds, and open markers at the budget are censored runs. Solid lines join the cell medians where at least 3 seeds grok. Dashed lines are the fitted plane of the main row of Table 4. The two grey lines at the lower right show the slopes 0 (rotation clock) and -1 (weight-decay clock).

Table 4: Fits of (5) at $\alpha = 1$ beside the predictions of Table 1. Brackets give 95% intervals from a bootstrap that resamples whole cells and then seeds within cells. The main row uses the 90 runs with $10^{-5} \le \eta\lambda \le 10^{-4}$, where every seed at every width groks. The other rows are checks.

Fit	a	b	γ	σ
Rotation clock predicts	> 0	≈ 0		
Weight-decay clock predicts	≈ 0	≈ -1		
Main range, level 0.95	$+0.24$ [$+0.08$, $+0.40$]	-0.63 [-0.92, -0.46]	-0.16 [-0.37, $+0.05$]	0.34
Adding $\eta\lambda = 3 \times 10^{-4}$	$+0.36$ [$+0.03$, $+0.62$]	-0.21 [-0.60, $+0.14$]	$+0.14$ [-0.24, $+0.48$]	0.80
Level 0.80	$+0.30$ [$+0.17$, $+0.45$]	-0.55 [-0.74, -0.39]	-0.17 [-0.36, -0.02]	0.29
Level 0.90	$+0.27$ [$+0.13$, $+0.41$]	-0.60 [-0.84, -0.42]	-0.16 [-0.37, $+0.03$]	0.31
Memorisation step	-1.53 [-1.80, -1.32]	$+0.07$ [-0.13, $+0.31$]	$+0.05$ [-0.23, $+0.29$]	0.38

Both exponents are nonzero, and both intervals exclude zero ($a = +0.24$, $b = -0.63$). A wider network groks later, as the rotation clock predicts, and more weight decay shortens the time, as the weight-decay clock predicts. Neither exponent takes the value its account predicts. Wider networks memorise sooner, so the width effect on grokking is not a delay in fitting the training data (memorisation row). The estimate of γ is negative, which would mean that weight decay shortens the time more at large width. Its interval includes zero, so the data do not fix its sign. Adding the cells at $\eta\lambda = 3 \times 10^{-4}$, where the widest networks stop grokking, changes the fit: b moves to -0.21 and its interval includes zero. The exponents therefore hold for the range in which every seed groks.

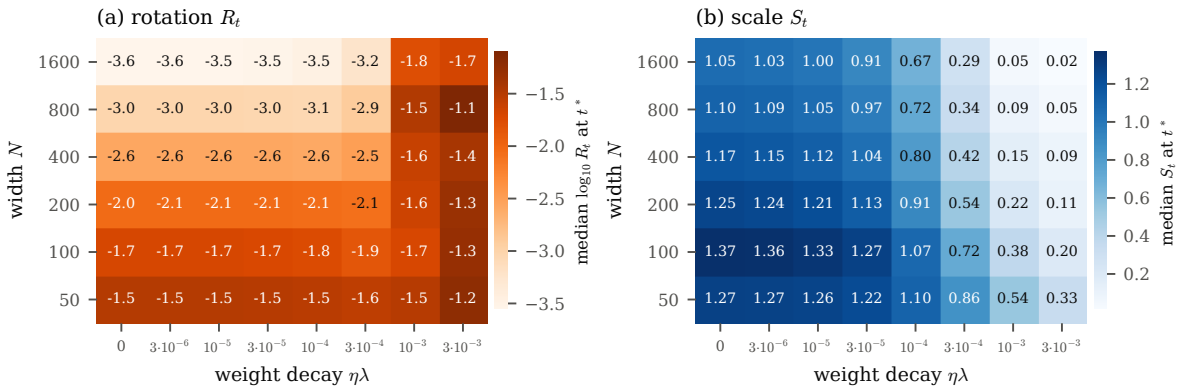


Figure 5: Width sets the rotation of the kernel, and weight decay shrinks its scale. Medians over seeds at $\alpha = 1$, read at $t^* = 2\,430$, before any run groks: (a) $\log_{10} R_t$ and (b) S_t.

Up to $\eta\lambda = 3 \times 10^{-4}$, the rotation at t^* depends on width and hardly on weight decay (Figure 5). Narrow networks rotate the most. The scale falls with weight decay at every width. Among the runs that grok, width alone explains 0.66 of the spread of $\log R_t$, and weight decay 0.18. The spread of S_t is shared: α explains 0.28, width 0.22 and weight decay 0.26. This links Tier 2 back to Tier 1. Width sets the rotation, weight decay is one of the settings that set the scale, and the grokking time depends on both.

The intervals in Table 4 resample cells as well as seeds. The misfit of the cells about the fitted plane has an SD of 0.32 in $\log T$, against 0.11 for the spread between the seeds of one cell. An interval that resamples seeds alone holds the cells fixed and is much narrower: for a it runs from +0.21 to +0.26.

4.4 Tier 3′: predicting the grokking time from the kernel

Page budget: 1.6 pages.

Claim. Read before any run groks, rotation is the best single predictor of the grokking time on held-out cells. Scale adds information that rotation lacks, and alignment adds little. The best models use rotation and scale together and let them interact.

The settings α and $\eta\lambda$ are the controls of this experiment. In a general network the laziness and the effective regularisation cannot be read off directly, while the tangent kernel can be measured. We therefore ask how much of the grokking time the kernel statistics recover on their own. A model that is given the settings serves as a reference that knows more than any network would report.

Table 5: Kernel ridge regression on subsets of the statistics at t^*, for the 355 runs that grok, scored with each of the 74 cells held out in turn. The last column is the share of the spread of each quantity that lies between cells.

Statistics	R^2	Concordance	Quantity	Between cells
A_t	0.05	0.55	$\log T$	0.96
S_t	0.27	0.67	S_t	0.99
$\log R_t$	0.35	0.70	$\log R_t$	0.99
S_t and $\log R_t$	0.51	0.76	A_t	0.56
$\log R_t$ and A_t	0.39	0.72		
S_t and A_t	0.23	0.65		
All three	0.55	0.78		

Taken alone, $\log R_t$ explains the most of the grokking time (Table 5). Adding S_t raises R^2 from 0.35 to 0.51, so rotation does not carry all of the information. Adding A_t to both raises it only to 0.55. The rotation and the scale interact. Growth of the kernel shortens the grokking time more when the kernel has also rotated more (Figure 7). In the censored model with all three statistics and their products, the coefficient of the product of S_t and $\log R_t$ is -0.62, with a cell-bootstrap interval from -1.39 to -0.09. Between 0.96 and 0.99 of the spread of $\log T$, S_t and $\log R_t$ lies between cells (Table 5). The statistics are therefore almost fully set by the cell, and they say nothing about which seed of a cell groks first.

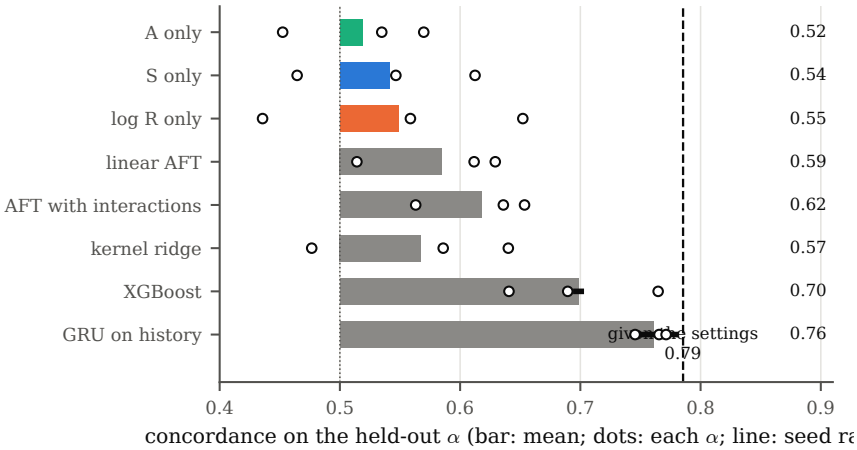


Figure 6: Predicting the grokking step of runs with an unseen α. Each model is fitted on two values of α and scored on the third by the concordance index with censoring. Bars are the mean over the three held-out values and dots each value. Black lines give the range over seeds (5 for the trees, 3 for the recurrent network). The dashed line is the trees given the settings N and $\eta\lambda$.

On an unseen α each statistic alone ranks runs only a little better than a coin toss (Figure 6): 0.52 for A_t, 0.54 for S_t and 0.55 for $\log R_t$. The linear model reaches 0.59 and the linear model with interactions 0.62. Kernel ridge reaches 0.57, gradient-boosted trees 0.70 and the recurrent network on the history 0.76. The reference model given the settings reaches 0.79, more than any model of the kernel statistics. Nanda et al. (2023, p. 1, arXiv v3) call a progress measure a metric that varies smoothly, precedes the phase transition and is causally linked to it. Rotation meets the first two conditions here. We have not tested a causal link, and reaching a rotation does not make a network grok (section 4.2). Rotation also gives no earlier or better warning than the settings themselves.

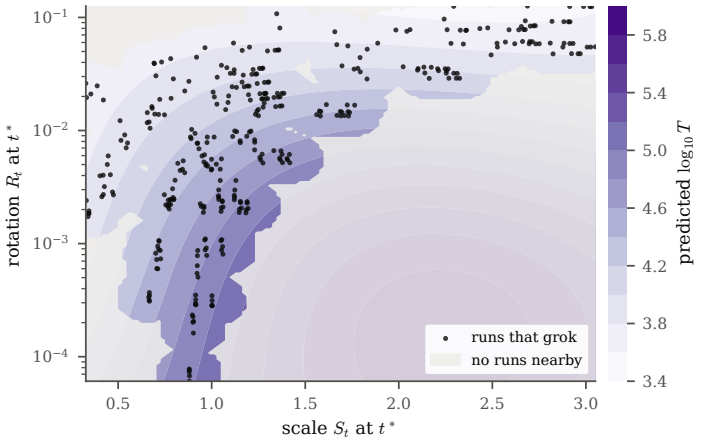


Figure 7: Growth of the kernel shortens the grokking time more when the kernel has also rotated more. The predicted grokking step over S_t and R_t at t^* comes from kernel ridge on the 355 runs that grok, with A_t at its median. Points are runs. The shaded region lies more than 0.25 standard deviations from every run, so the surface there is an extrapolation.

4.5 Tier 4: an extension to MNIST digits

[TODO: This section will be populated with an extension experiment on MNIST digits.]

5 Limitations

Page budget: 0.6 page.

[TODO: Write each point as one or two sentences.]

- Every run uses the NTK parameterisation, one task and one architecture. A width effect measured here belongs to this parameterisation.
- Each cell has 5 seeds.
- The events are known only to the resolution of the checkpoint grid.
- The fits in Tier 2 depend on which cells enter, and the decay edge bends the plane (Table 4).
- The kernel statistics say nothing about which seed of a cell groks first, and the reference model given the settings predicts better than any kernel model.
- The alignment here uses the kernel of the summed logits. The tangent kernel of all p logits may behave differently.
- The centred predictor is not homogeneous, so the theorem behind (1) applies to our network only as a guide.
- The proposal also planned a clamped-norm arm, a sweep over p and the training fraction, and a cross-entropy arm. They were not run.
- The proposal defined the events by the loss. The report uses accuracy (section 3.4).

6 Conclusion

Page budget: 0.3 page.

We asked what sets the grokking time: the shape of the tangent kernel, its scale, or the weight-decay rate. Rotation matters most and is not enough on its own. At a fixed width it is the statistic most tied to the grokking step, yet networks can reach that rotation without grokking. The grokking time grows with width and falls with weight decay. Width sets the rotation, and at a fixed width weight decay shrinks the scale. Early rotation and scale together predict the grokking time better than either one alone, so each published clock describes one part of it. What is new is the split into scale and rotation, the placebo test, the crossed grid, and prediction from quantities that any network exposes. A next step is to test whether the early rate of rotation falls as $1/N$.

Contribution

[TODO: One line for each member, from the group.]

- Al-Saffi: **[TODO:]**
- Chong: **[TODO:]**
- D'Mello: **[TODO:]**
- Kienzle: **[TODO:]**
- Rieger: **[TODO:]**

Acknowledgement

[TODO: Anyone outside the group who advised on the project.]

Use of AI tools

[TODO: The task sheet asks each deliverable to state how AI tools were used. The group should agree on this text. A draft follows.] The group used an AI assistant (Claude, by Anthropic) to draft parts of the code, to restructure this report, and to check references against their arXiv pages. Group members checked every result and every reference before submission.

A The identity for the variation

Page budget: 0.5 page (appendices: 5 pages in all).

Expanding the squared norm gives $\|K_t - K_0\|_F^2 = \|K_t\|_F^2 - 2\langle K_t, K_0\rangle_F + \|K_0\|_F^2$. Dividing by $\|K_0\|_F^2$ and using $\langle K_t, K_0\rangle_F = (1-R_t)\|K_t\|_F\|K_0\|_F$ from (3) gives $D_t^2 = S_t^2 - 2S_t(1-R_t)+1$. By the Cauchy-Schwarz inequality, $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$, and then $D_t = |S_t - 1|$. **[TODO: Check against the golden report, Appendix A.1.]**

B The baseline against Kumar et al.

Page budget: 0.75 page.

Table 6 sets each setting of our baseline beside Kumar et al. (2024, arXiv v3). We read their Appendices 8.1 and 8.3, Section 3 and Figure 2. They do not state the parameterisation or the initialisation. Table 7 sets our event steps beside the steps read by eye from their Figure

2, which shows one run for each α. Their grokking at $\alpha = 0.5$ and $\alpha = 1$ falls close to ours. Their training accuracy rises much more slowly. At $\alpha = 1$ it reaches 100% near 20 000 epochs, while our runs reach 0.99 by about 3 000 steps. They also sweep $\alpha = 1.5$, which our grid leaves out.

Table 6: The baseline setting by setting. "Where" is the place in Kumar et al. (2024, arXiv v3) that gives the setting.

Setting	Kumar et al. (2024)	This report	Where
Task	$a + b \bmod p$, $p = 23$	$a + b \bmod p$, $p = 23$	App. 8.3
Encoding	two one-hot codes, $D = 2p$	two one-hot codes, $D = 2p$	App. 8.3
Output	p outputs, one-hot target	p outputs, one-hot target	App. 8.3
Training pairs	90% of p^2	90% of p^2 (476 of 529)	App. 8.3
Width	$N = 100$	$N = 100$ (grid: 50 to 1600)	App. 8.3
Network	one hidden layer	one hidden layer, no biases, rectifier	App. 8.3
Parameterisation	not stated	NTK, weights from $\mathcal{N}(0, 1)$	
Loss	mean squared error	mean squared error	Sec. 3
Optimiser	vanilla full-batch GD, no weight decay	full-batch GD; $\eta\lambda = 0$ at the baseline	App. 8.3
Predictor	$\alpha[f(x;\theta) - f(x;\theta_0)]$	the same	App. 8.1
Learning rate	$\eta = 100$; $\eta = \eta_0/\alpha^2$	$\eta = 100/\alpha^2$	App. 8.1, 8.3
α	$\{0.5, 1, 1.5, 2\}$	$\{0.5, 1, 2\}$	Fig. 2(c)
Budget	100 000 epochs (Fig. 2(c))	200 000 steps	Fig. 2(c)
Seeds	one run shown	5 per cell	Fig. 2

Table 7: Event steps at $N = 100$ and $\eta\lambda = 0$. The values of Kumar et al. (2024) are read by eye from their Fig. 2(a) and (c), so they are approximate.

	Kumar et al. (2024), read from Fig. 2		This report, median [range] over 5 seeds	
α	Train 100%	Test 100%	Memorisation	Grokking (0.95)
0.5	≈4 000	≈5 000	1 455 [1 313, 1 612]	4 981 [3 662, 6 776], 5/5
1	≈20 000 to 25 000	≈25 000 to 27 000	2 983 [2 692, 2 983]	23 000 [15 395, 34 000], 5/5
1.5	≈65 000	≈70 000	not in the grid	
2	≈90 000	about 25% at 100 000	5 520 [4 496, 6 000]	0/5 by 200 000

C Outcomes over the whole grid

Page budget: 1.0 page.

Table 8 gives the outcome of every cell. An entry gives the seeds that grok at level 0.95 out of 5. After a colon it gives the median grokking step in thousands, when at least 3 seeds grok. A cell in which no seed groks shows 0 and the most common phase at the last step, as in Figure 3. Every seed groks in 35 of the 48 cells at $\alpha = 0.5$, 26 at $\alpha = 1$ and 8 at $\alpha = 2$.

Table 8: Outcomes over the whole grid at level 0.95: seeds that grok out of 5, and the median grokking step in thousands.

α	N	0	3×10^{-6}	10^{-5}	3×10^{-5}	10^{-4}	3×10^{-4}	10^{-3}	3×10^{-3}
0.5	50	5: 7	5: 7	5: 8	5: 7	5: 7	5: 5	5: 6	0 N
	100	5: 5	5: 5	5: 4	5: 4	5: 4	5: 4	5: 4	0 N
	200	3: 27	5: 26	5: 14	5: 7	5: 5	5: 4	5: 6	0 N
	400	0 M	5: 70	5: 19	5: 9	5: 6	5: 4	0 M	0 N
	800	0 M	5: 106	5: 33	5: 12	5: 7	5: 6	0 N	0 N
	1600	0 M	5: 157	5: 48	5: 17	5: 8	5: 9	0 N	0 N
1	50	5: 33	5: 30	5: 27	5: 24	5: 19	5: 15	0 M	0 N
	100	5: 23	5: 23	5: 21	5: 17	5: 10	5: 9	0 M	0 N
	200	0 M	0 M	5: 106	5: 27	5: 11	5: 10	0 N	0 N
	400	0 M	0 M	5: 82	5: 27	5: 12	5: 14	0 N	0 N
	800	0 M	0 M	5: 88	5: 30	5: 15	0 G	0 N	0 N
	1600	0 M	0 M	5: 99	5: 37	5: 20	0 F	0 N	0 N
2	50	0 M	0 M	1	2	1	0 M	0 N	0 N
	100	0 M	0 M	3: 95	5: 69	5: 43	0 M	0 N	0 N
	200	0 M	0 M	0 M	5: 86	5: 40	0 M	0 N	0 N
	400	0 M	0 M	0 M	5: 83	5: 60	0 N	0 N	0 N
	800	0 M	0 M	0 M	5: 94	0 G	0 N	0 N	0 N
	1600	0 M	0 M	0 M	5: 98	0 M	0 N	0 N	0 N

D A cross-check from a second pipeline

Page budget: 0.75 page.

[TODO: One paragraph and one figure. A second training pipeline (release grid-v0.1.0 on branch feat/grid) ran 12 seeds per cell at the baseline with weight decays near the edge, with a budget of 100 000 steps. The golden report gives these results: the grokking time is flat in $\eta\lambda$ below a break at $\eta\lambda \approx 7.0 \times 10^{-4}$ [E14], rises steeply above it, and no seed groks at 8.28×10^{-4} [E14]. Export the figure as a PDF from that pipeline (E14).]

E Settings of the prediction models

Page budget: 1.0 page.

All models use the 630 runs with $\eta\lambda \leq 10^{-3}$, with S_t, $\log R_t$ and A_t read at t^*. The grokking step of a censored run is known only to exceed 200 000. Whole groups are held out: one value of α at a time (Figure 6) or one cell at a time (Table 5). The file `report_v2/code/models.py` fixes every setting below.

Linear models. The log-normal accelerated failure time models standardise the statistics on the training fold and add the products of the standardised statistics when they include interactions. They use a ridge penalty of 0.01 in cross-validation. The fit for the product coefficient of section 4.4 uses all runs, a penalty of 10^{-4}, and 200 bootstrap draws of whole cells with seed 0.

Kernel ridge. The kernel is Gaussian on standardised inputs. Its penalty in $\{0.01, 0.1, 1\}$ and its width in $\{0.1, 0.5, 2\}$ are chosen by grouped 5-fold cross-validation over the training cells, inside every outer fold. It is fitted to $\log T$ on the training runs that grok.

Gradient-boosted trees. The trees use the accelerated failure time objective with normal errors of scale 1 and a subsample of 0.8. The search covers depths 1 to 4, learning rates 0.03 and 0.1, and minimum child weights 1, 5 and 10, with up to 1 000 rounds and early stopping after 50. The setting with the best concordance on 20% of the training cells is refitted on all training runs. The seeds are 0 to 4. The reference model is the same search on $\log N$ and $\log_{10}(\eta\lambda)$, with $\eta\lambda = 0$ set to 10^{-6}.

Recurrent network. One gated recurrent layer with 16 units reads A_t, S_t and R_t at the 25 saved steps up to t^*. R_t enters without a logarithm because it is 0 at step 0. The inputs are standardised on the training fold. A linear head gives the location and a learned parameter the scale of a log-normal accelerated failure time loss. Training uses Adam with learning rate 0.01 and weight decay 10^{-4}, on the full batch, for up to 800 epochs. Every 10 epochs the loss on 20% of the training cells is checked, and the best weights are kept. Early stopping chose between 70 and 350 epochs. The seeds are 0 to 2.

F Sensitivity to the grokking level

Page budget: 0.5 page.

Table 9 repeats the placebo test at the levels 0.80 and 0.90, and Table 10 repeats the main fit. At $N = 100$ the order of the statistics is the same at every level. R_t tightens most, with ratios of 1.83 at 0.80 and 1.79 at 0.90, and S_t comes next. Across widths R_t never passes. At 0.80 the scale and the alignment tighten slightly across widths, with p of 0.024 and 0.001 and ratios of only 1.25 and 1.08. At 0.90 only the alignment does ($p = 0.013$); the scale has $p = 0.085$. The exponents a and b keep their signs at every level, and both intervals exclude zero.

Table 9: The placebo test of Table 3 at three grokking levels. The number of runs that grok is in brackets.

Level	Runs	Statistic	SD at grokking	Placebo SD	Ratio	p
0.80	$N = 100$ (86)	S_t	1.1	1.59	1.45	0.000
		R_t	0.0122	0.0223	1.83	0.000
		A_t	0.0291	0.033	1.13	0.002
0.80	$\alpha = 1$ (136)	S_t	1.01	1.26	1.25	0.024
		R_t	0.0376	0.0391	1.04	0.131
		A_t	0.0446	0.048	1.08	0.001
0.90	$N = 100$ (79)	S_t	1.18	1.58	1.34	0.000
		R_t	0.0118	0.021	1.79	0.000
		A_t	0.0307	0.0335	1.09	0.029
0.90	$\alpha = 1$ (134)	S_t	1.15	1.35	1.17	0.085
		R_t	0.0392	0.0403	1.03	0.195
		A_t	0.0475	0.0501	1.06	0.013
0.95	$N = 100$ (78)	S_t	1.25	1.6	1.29	0.000
		R_t	0.0119	0.0203	1.71	0.000
		A_t	0.0313	0.0341	1.09	0.029
0.95	$\alpha = 1$ (130)	S_t	1.33	1.44	1.08	0.258
		R_t	0.0408	0.0411	1.01	0.423
		A_t	0.0504	0.0521	1.03	0.079

Table 10: The main fit of Table 4 at three grokking levels, with intervals that resample cells and seeds.

Level	a	b	γ	σ
0.80	+0.30 [+0.17, +0.45]	−0.55 [−0.74, −0.39]	−0.17 [−0.36, −0.02]	0.29
0.90	+0.27 [+0.13, +0.41]	−0.60 [−0.84, −0.42]	−0.16 [−0.37, +0.03]	0.31
0.95	+0.24 [+0.08, +0.40]	−0.63 [−0.92, −0.46]	−0.16 [−0.37, +0.05]	0.34

References

Chen, T. and C. Guestrin (2016). *XGBoost: A Scalable Tree Boosting System*. DOI: 10.48550/arXiv.1603.02754. arXiv: 1603.02754 [cs.LG].

Chizat, L., E. Oyallon, and F. Bach (Jan. 2020). *On Lazy Training in Differentiable Programming*. DOI: 10.48550/arXiv.1812.07956. arXiv: 1812.07956 [math.OC].

Cho, K. et al. (2014). *Learning Phrase Representations Using RNN Encoder-Decoder for Statistical Machine Translation*. DOI: 10.48550/arXiv.1406.1078. arXiv: 1406.1078 [cs.CL].

Cortes, C., M. Mohri, and A. Rostamizadeh (Apr. 2024). *Algorithms for Learning Kernels Based on Centered Alignment*. DOI: 10.48550/arXiv.1203.0550. arXiv: 1203.0550 [cs.LG].

Davidson-Pilon, C. (Aug. 2019). "Lifelines: Survival Analysis in Python". In: *Journal of Open Source Software* 4.40, p. 1317. ISSN: 2475-9066. DOI: 10.21105/joss.01317.

Fort, S. et al. (Oct. 2020). *Deep Learning versus Kernel Learning: An Empirical Study of Loss Landscape Geometry and the Time Evolution of the Neural Tangent Kernel*. DOI: 10.48550/arXiv.2010.15110. arXiv: 2010.15110 [cs.LG].

Geiger, M. et al. (Nov. 2020). "Disentangling Feature and Lazy Training in Deep Neural Networks". In: *Journal of Statistical Mechanics: Theory and Experiment* 2020.11, p. 113301. ISSN: 1742-5468. DOI: 10.1088/1742-5468/abc4de.

Gromov, A. (Jan. 2023). *Grokking Modular Arithmetic*. DOI: 10.48550/arXiv.2301.02679. arXiv: 2301.02679 [cs.LG].

Harrell Jr., F. E. et al. (1982). "Evaluating the Yield of Medical Tests". In: *JAMA* 247.18, pp. 2543–2546. DOI: 10.1001/jama.1982.03320430047030.

Jacot, A., F. Gabriel, and C. Hongler (Feb. 2020). *Neural Tangent Kernel: Convergence and Generalization in Neural Networks*. DOI: 10.48550/arXiv.1806.07572. arXiv: 1806.07572 [cs.LG].

Kalbfleisch, J. D. and R. L. Prentice (2002). *The Statistical Analysis of Failure Time Data*. 2nd ed. Wiley Series in Probability and Statistics. Wiley. ISBN: 978-0-471-36357-6. DOI: 10.1002/9781118032985.

Khanh, T. X. (June 2026). *What Does the Weight Norm Control in Grokking? Logit-scale Mediation under Cross-Entropy*. DOI: 10.48550/arXiv.2606.18465. arXiv: 2606.18465 [cs.LG].

Khanh, T. X. et al. (June 2026). *The Weight Norm Sets the Grokking Timescale: A Causal Delay Law*. DOI: 10.48550/arXiv.2606.13753. arXiv: 2606.13753 [cs.LG].

Kim, T. (July 2026). *Grokking on the Weight-Decay Clock: A Rate Hierarchy from Softly Broken Symmetries*. DOI: 10.48550/arXiv.2607.23967. arXiv: 2607.23967 [cs.AI].

Kumar, T. et al. (Apr. 2024). *Grokking as the Transition from Lazy to Rich Training Dynamics*. DOI: 10.48550/arXiv.2310.06110. arXiv: 2310.06110 [stat.ML].

Lewkowycz, A. and G. Gur-Ari (Jan. 2021). *On the Training Dynamics of Deep Networks with l_2 Regularization*. DOI: 10.48550/arXiv.2006.08643. arXiv: 2006.08643 [stat.ML].

Mohamadi, M. A., W. Bae, and D. J. Sutherland (2022). *A Fast, Well-Founded Approximation to the Empirical Neural Tangent Kernel*. DOI: 10.48550/arXiv.2206.12543. arXiv: 2206.12543 [stat.ML].

Mohamadi, M. A. et al. (July 2024). *Why Do You Grok? A Theoretical Analysis of Grokking Modular Addition*. DOI: 10.48550/arXiv.2407.12332. arXiv: 2407.12332 [cs.LG].

Nanda, N. et al. (Oct. 2023). *Progress Measures for Grokking via Mechanistic Interpretability*. DOI: 10.48550/arXiv.2301.05217. arXiv: 2301.05217 [cs.LG].

Power, A. et al. (Jan. 2022). *Grokking: Generalization beyond Overfitting on Small Algorithmic Datasets*. DOI: 10.48550/arXiv.2201.02177. arXiv: 2201.02177 [cs.LG].

Pracher, L. et al. (Sept. 2026). *A Spectral Theory of Grokking: Weight Decay Induces Feature Learning*. DOI: 10.48550/ARXIV.2609.26679. arXiv: 2609.26679 [cs.LG].

Indicate whether all your group members consent for your report to be used as a teaching resource by including one of the statement below.

We give consent for this to be used as a teaching resource.
We DO NOT give consent for this to be used as a teaching resource.

[TODO: Keep one of the two statements.]

Draft notes: delete before submission

1. How the report meets each marking criterion

The task sheet (`project.pdf`) marks the report on five criteria and asks the main text to cover six points within 12 pages. The table gives where the report addresses each one as it stands on 12 October 2026, the mark we expect, and how to raise it. The marks are our own estimate; the task sheet gives no scale.

Criterion	Where it is addressed	Likely mark	How to improve
Clarity and conciseness. No spelling, grammar or punctuation mistakes; well organised; clear notation; straight to the point.	One research question. Three statistics on one kernel, defined in Equation. (3). Each section of section 4 opens with its claim. Seven figures and five tables in the main text, with one colour for each statistic in every figure. The spectral and survival machinery is cut.	Good.	Write the Limitations as short prose; it is still a list under a TODO. Decide on the notation table. Do one pass for spelling and for the writing rules of CLAUDE.md, and one read-through by a member who did not write the section.
Correctness. Sound arguments and technical details.	Censored log-normal fits whose intervals resample cells, with the check rows of Table 4. The placebo test with 2 000 permutations. Grouped cross-validation by cell and by α, with a settings reference. Every claim is held to the limits in `report_v2/CLAUDE.md`, and claims the full runs did not support were softened on 12 October (changelog).	Good.	Add one check of the log-normal assumption, such as a probability plot of the residuals of the main fit; it was cut with the golden report's survival material. Settle the open reference items: Truong or Khanh, the Kalbfleisch edition, and Dyer and Gur-Ari.
Reproducibility. Enough detail to reproduce the results.	Table 2 gives every setting, the split rule and the compute. Appendix E gives every setting of the prediction models. `report_v2/code/` rebuilds every figure, table and number with `make_all.py` from a 9 MB extract with checksums. Its README maps each item to its script, and every number in the text is a macro.	Good.	Run E14 or drop Appendix D; it is still a TODO. Package `report_v2/code/` as the zip for Blackboard, with the data extract, and name in the report where the code is.

Criterion	Where it is addressed	Likely mark	How to improve
Significance. The report should highlight what it believes is significant.	The Introduction sets the question against three 2026 accounts that disagree. Related Work ends with what each account predicts (Table 1). Tier 1 offers the placebo check to other threshold claims. Tier 2 gives a practical point: the weight decay that speeds grokking depends on the width. The Conclusion states the answer.	Medium to good.	State what a single small task can and cannot show. Say in one sentence why a practitioner would measure S_t and R_t, since the settings reference still predicts better (Figure 6).
Novelty. The report should highlight what it believes is novel.	A paragraph after the contributions in the Introduction, and the Conclusion: the split of the kernel's movement into scale and rotation as timing variables, the placebo test, the crossed grid of width and weight decay, and prediction from measurable quantities. No sentence claims a first.	Medium to good. The split may look small.	Search the literature beyond the works cited before any claim of a first. Contrast the split with the distance of Mohamadi et al. (2024) more sharply in section 4.1.
Required content. Problem, significance, steps, findings, limitations and conclusion within 12 pages; contribution; AI use; consent.	Problem and significance in Sections 1 and 2, steps in Section 3, findings in Section 4, limitations in Section 5 and conclusion in Section 6. Contribution, AI use and consent sections exist.	Nearly complete.	Without the red markup the main text ends two lines into page 13; cut them (see the changelog of 12 October). The appendices take about 4 pages of the 5 allowed. Fill in the contribution lines, the acknowledgement and the consent choice, and agree the AI-use text. Name the file G2_report_<Name>.pdf and submit the code separately as a zip.

2. Changelog: decisions and what was cut

Decisions taken on 5 October 2026.

- All results come from `data/`: 720 runs, 5 seeds, 200 000 steps, $\alpha \in \{0.5, 1, 2\}$, six widths and eight weight decays. The golden report used the grid-v0.1.0 databases (12 seeds, 100 000 steps) instead.
- The alignment A_t is the centred alignment of the summed-logit kernel on the test pairs (`A_sum` in `data/`). The golden report on main uses the tangent kernel of all p logits, after Baratin et al. The statistics S_t, R_t and A_t now share one kernel, and the prediction notebooks already use this A_t.
- The main text uses one grokking level, 0.95. The levels 0.80 and 0.90 move to Appendix F.

- The reproduction of Kumar et al. (2024) keeps a short section with one figure. The comparison setting by setting moves to Appendix B.
- Tier 4′ was first drafted as the rate-addition model. It was reopened the same day as a decision between five options (A to E), because that model rests on assumptions of our own.
- Each section gained a red draft contribution block that states, as if the preliminary findings hold, what is new and why it matters.
- The word "rotation" is kept for now and will become "shape" later.
- In Tier 3′ the model given the settings is a reference, since a general network does not report its laziness or its effective regularisation.
- The planning checks changed one claim. The alignment does not pass the placebo test, so Tier 1 claims a threshold for rotation only.

Decisions taken on 11 October 2026 (Jasper Chong).

- Tier 4′ (Section 4.5, with its options A to E and its figure) is commented out in `main.tex`, and items E11, E17 and E18 were not run. To restore it, remove the leading comment sign from each line between the markers TIER 4' COMMENTED OUT and END OF TIER 4', and restore contribution 5 of the Introduction.
- (12 October 2026, Jasper Chong.) A new Section 4.5, Tier 4, holds a TODO placeholder for an extension experiment on MNIST digits. It is separate from the commented-out theory section Tier 4′. The MNIST runs would be new training outside `data/`.
- The spread in Tier 1 is the standard deviation, not the coefficient of variation. To restore the CV, set `SPREAD = 'cv'` in `report_v2/code/E03_placebo.py`; its JSON output keeps the CV values.
- Every figure, table and number now comes from the scripts in `report_v2/code/` through `numbers.tex`, `figures/` and `tables/`. The red planning values are gone except those of E14.
- The Tier 2 intervals resample whole cells and then seeds within cells. This settles the open decision on the intervals. Seed-only intervals are in `E06.json` for comparison.
- E14 (Appendix D) was not rerun. It needs the grid-v0.1.0 databases and the R scripts of branch `feat/grid`.
- Kumar et al. App. 8.1 and 8.3, Sec. 3 and Fig. 2 (arXiv v3) were read for Appendix B. The memorisation steps differ from their figure; the grokking steps agree at $\alpha = 0.5$ and $\alpha = 1$.
- Claims changed by the full runs were marked in red with "Differs from draft" in the draft contribution blocks and the abstract. In short: the rotation threshold at a fixed width is not sufficient; the interval of γ includes zero; weight decay is one of the settings that set the scale; across widths no statistic passes only at level 0.95. These markers were removed on 12 October 2026 (below).

Changes made on 12 October 2026.

- The red draft contribution blocks were folded into the prose and deleted, together with the `draftcontrib` environment, the `\differs` macro and the commented block inside Tier 4′. Their text is in the git history of `main.tex` before this date. The abstract and the Conclusion were written from the final numbers.

- The "why it matters" and "what is new" points that the numbers support now sit in the Introduction (a paragraph after the contributions), Related Work (the Predictions paragraph and Kim's scope), section 4.1 (the gate and the baseline split), section 4.2 (the alignment hypothesis and the critical norm), section 4.3 (Kim's scope and the width-dependent decay edge) and section 4.4 (scoring, progress measure).
- Claims changed. (1) The Tier 2 claim and the Conclusion said that weight decay acts through the scale. Weight decay explains 0.26 of the spread of S_t, about as much as α and width, so the text now says that at a fixed width weight decay shrinks the scale. The story in `report_v2/CLAUDE.md` still says weight decay sets the scale. (2) The baseline claim "rotates by a few per cent" became "R_t stays below 0.09", the bound on every baseline checkpoint. (3) The reconciling claim of the Introduction block was dropped.
- Claims added from the final numbers. Tier 1 now reports that S_t pooled over all runs that grok tightens 1.69-fold, with the caveat that it keeps growing. Tier $3'$ now states in the prose that 0.96 to 0.99 of the spread lies between cells, so the statistics do not separate the seeds of a cell, and its claim is limited to held-out cells.
- The Introduction no longer carries a "search the literature" TODO. No sentence claims a first. The search is listed under novelty in the criteria table.
- The criteria table was rewritten against the five criteria and the required content of the task sheet.
- The red TODO around the interval sentence of section 3.5 was removed, since the intervals resample cells (decision of 11 October).
- Page limit. Built with the red markup hidden, the main text ended five lines into page 13. Words that repeat other sections were cut: Kim's scope in section 4.3 (kept in Related Work), a closing sentence of section 4.1, a scoring sentence of section 4.4 (kept in section 3.5), and parts of the Introduction and Conclusion. Fig. 7 went from 0.66 to 0.6 of the line width. Two lines still spill onto page 13. The next cuts would be to write the Limitations as prose, or to merge Figure 5 into Figure 3.
- The code and the numbers were verified. The tangent kernel recomputed by autograd from the saved weights of 8 runs matches S_t, R_t and A_t to a relative 3×10^{-10}. A minimal retraining of the baseline run reproduces its accuracy curve exactly over 6 000 steps. Independent code reproduces 133 macros of `numbers.tex`; the only gaps are Monte Carlo ones (a permutation p and a bootstrap bound, both within their noise). A clean rerun of `make_all.py` reproduces `numbers.tex`, every table and every JSON output exactly.
- One claim was corrected. The text said that at levels 0.80 and 0.90 the scale and the alignment tighten across widths. At 0.90 the scale has $p = 0.085$, so only the alignment does. Section 4.2 and appendix F now say so.

Cut from the golden report. Each item can be restored from the golden report's source, `docs/report_golden.tex` (added in commit 7da1996), and its figure files in `docs/figures/`.

- Section 3.4, Spectral measures: effective rank, trace ratios and the target power along eigenvectors. Section 4.3: the cumulative target power figure (`docs/figures/spectrum.tex`).
- Section 3.5, Event times: the Kaplan-Meier estimate with Greenwood's interval, the phase labels, the additive model of Aalen and the relative risk model with time-dependent covariates. Section 4.3: the table of that model. Appendix B, Model checks: the probability plot and the cumulative regression functions (`docs/figures/probability.tex`, `aalen.tex`).

- Section 3.6, Choice of levels: optimal designs, the censored information matrix, the broken line and the sequential design. Section 4.2.2: the scan, the design table, the broken-line figure, the survival figure and the accelerated failure time table of the weight-decay arm (`docs/figures/phases.tex`, `broken.tex`, `survival.tex`). A short summary stays in Appendix D.
- Section 4.2.1: the α sweep with $\alpha = 1.5$ and 12 seeds (`docs/figures/alpha_sweep.pdf`).
- Section 3.2: the first-logit kernel as a second output.
- Related Work: the paragraph on kernel alignment and spectral bias, silent alignment, the early and late dichotomy, Omnigrok, and provable grokking in ridge regression.
- The pipeline itself: branch `feat/grid`, folder `grid/` (Snakemake workflow, DuckDB databases, R scripts).

Planned in the proposal and not run. The clamped-norm arm with $\rho \in [1, 2]$, the arm over three primes and three training fractions, the cross-entropy arm, the statistic $y^\top K^{-1} y$, and events defined by loss thresholds. The proposal is `docs/G2_proposal.tex`.

Superseded analyses. `repoduced-code/tier1_v2.ipynb` and `tier2_width_decay.ipynb` use the old data, with one split shared by every seed, the trace kernel and test accuracy 1.0. Their methods are ported, and their numbers are not used.

Open decisions for the group.

- Keep the title, or change it to give the answer.
- Keep Figure 5 as its own figure, or merge it into Figure 3.
- Whether to add the notation table.
- Which theory direction, if any, Tier 4′ takes (options A to E in the commented-out Section 4.5 of `main.tex`).
- Confirm the softer Tier 2 headline of 12 October 2026 ("at a fixed width, weight decay shrinks the scale"), and update the story in `report_v2/CLAUDE.md` to match.

3. Reference checks

Every entry of `ref.bib` was checked on 5 October 2026 against its arXiv abstract page, its DOI record on Crossref, or PubMed. Each section and equation number cited in the text was checked in the PDF of the version named.

Entry	Checked against	Status	Notes
Power et al. 2022	arXiv 2201.02177	Matches	v1 only.

Entry	Checked against	Status	Notes
Nanda et al. 2023	arXiv 2301.05217	Matches	v3 of 19 Oct 2023 gives the month. The v3 PDF was read on 12 Oct 2026 for the progress-measure sentence of section 4.4. Page 1 defines progress measures as "metrics that precede and are causally linked to the phase transition, and which vary more smoothly", after Barak et al. (2022). The report claims only the first and last conditions for R_t.
Gromov 2023	arXiv 2301.02679	Matches	
Jacot et al. 2020	arXiv 1806.07572	Matches	Year is v4 (Feb 2020); NeurIPS 2018.
Chizat et al. 2020	arXiv 1812.07956	Matches	Year is v5 (Jan 2020); NeurIPS 2019.
Kumar et al. 2024	arXiv 2310.06110	Matches	v3 of 11 Apr 2024. App. 8.1 and 8.3, Sec. 3 and Fig. 2 read in the v3 PDF on 11 Oct 2026 (Appendix B).
Mohamadi et al. 2024	arXiv 2407.12332	Matches	ICML 2024.
Mohamadi, Bae, Sutherland 2022	arXiv 2206.12543	Matches, entry type changed	The golden bib types it as an article with no journal. It is typed misc here. Published at ICML 2023.
Lewkowycz, Gur-Ari 2021	arXiv 2006.08643, PDFs of v1 and v2	Matches	Year is v2 (Jan 2021); NeurIPS 2020. CLAUDE.md gives both 2020 and 2021. The kernel flow is Eq. S5 and T_t is Eq. S6 in both v1 and v2. The proposal's draft note, which says they become S8 and S9 in v2, is wrong: S8 in v2 is the infinite-width solution. $E[T_0] = O(n^{-1})$ is credited to Dyer and Gur-Ari (2020), which is not in ref.bib.
Fort et al. 2020	arXiv 2010.15110	Matches	NeurIPS 2020.
Geiger et al. 2020	Crossref and arXiv 1906.08034	Matches	JSTAT 2020, 113301.
Cortes et al. 2024	arXiv 1203.0550	Matches	Year is v3 (Apr 2024). The paper is JMLR 13 (2012), 795–828.
Khanh et al. 2026	arXiv 2606.13753	Matches arXiv; name order flagged	arXiv lists "Truong Xuan Khanh" and parses the surname as Khanh, so "Khanh et al." matches arXiv. In Vietnamese order the family name is usually written first, which would make it "Truong et al.". The co-authors have the same question. The proposal's key is `truong2026norm`. The group should decide.
Khanh 2026	arXiv 2606.18465	Matches arXiv; name order flagged	As above.

Entry	Checked against	Status	Notes
Kim 2026	arXiv 2607.23967	Matches	Cor. 5 and Eq. 20 checked in the v1 PDF (page 8). The Scope paragraph cited in Section 2 and Tier 2 is on page 3 of v1. It lists grokking without explicit weight decay and the creation of new population-visible directions by feature learning among what the mechanism does not explain.
Pracher et al. 2026	arXiv 2609.26679	Matches, fields added	The golden bib gives only the DOI. The eprint, month and class were added. v1 of 22 Sep 2026.
Kalbfleisch, Prentice 2002	Crossref, Open Library	Matches, edition changed	The golden bib gives edition 1. The 2002 Wiley printing is the second edition; the first appeared in 1980. Crossref lists the online version as edition 1, so this needs a check of the book itself.
Davidson-Pilon 2019	Crossref	Matches	JOSS 4(40), 1317.
Harrell et al. 1982 (new)	PubMed 7069920, Crossref	Matches	JAMA 247(18), 2543–2546. Crossref lists only the first author; PubMed lists all five.
Chen, Guestrin 2016 (new)	arXiv 1603.02754	Matches	KDD 2016.
Cho et al. 2014 (new)	arXiv 1406.1078	Matches	EMNLP 2014.